Instructions

There are **50 total points**. This exam is closed-book and closed-notes except for **one 8.5 × 11-inch sheet of paper with hand-written notes (both sides permitted)**. No electronic devices, additional pages, or other materials are permitted. When asked to provide your answer within a figure or table, be careful to not exceed box boundaries. Bubbles must be filled out completely: ● is correct, ✓ ⊙ ☒ are incorrect . Write your full name in the box to acknowledge the instructions.

Brian Jiang

Motivation and Applications

1. [3 points]: Since most streaming video became encrypted, which of the following can a network operator *no longer read directly* from the traffic, and so must infer? Select all that apply.

- [x] Video resolution
- [x] Rebuffering (stall) events
- [x] Startup delay before playback
- [ ] Packet sizes and timing
- [ ] IP addresses of the video servers
- [ ] Number of parallel connections

2. [2 points]: In Hands-On 1 you looked at a packet capture before doing anything else with it. Which observations should make you suspect a problem with the data? Select all that apply.

- [x] A packet with a negative length
- [x] A packet far above the maximum packet size
- [x] Timestamps that go backwards
- [ ] Large packets flow mostly server to client
- [ ] A handshake before the large packets
- [ ] Round-trip times that vary over the trace

3. [3 points]: Attackers are preparing a phishing campaign that imitates the University's login page. You run the campus network and can obtain domain registration data and the logs of the campus DNS resolver. Name **one** data source and **two** features you would use to detect the campaign, ideally before the first email arrives.

(Answer inside the box)

Data source: flow capture / packet logs on top of WHOIS and DNS
Features: small edit distance to brand (the university login page here), new domain (newer domains may have been created for the attack)

4. [3 points]: In Hands-On 2, the Log4j scan and the normal web trace were captured at different times of day, so a model with time of day as a feature separates them perfectly. Is time of day a good feature for detecting scans? ○ Yes ● No

Why or why not?

(Answer inside the box)

Log4j and normal web traffic can happen at any time of day, a model with time of day as a feature would not generalize well to other traces

Data Acquisition

5. [3 points]: Which of the following are *passive* measurements? Select all that apply.

- [x] An ISP collects flow records from its routers
- [x] Campus logs every query its DNS resolver answers
- [x] Inferring resolution from a capture of your Netflix session
- [ ] An app downloads a test file every hour
- [ ] A laptop pings a server once a minute
- [ ] Measuring latency while a test upload runs

6. [3 points]: A student reports that Zoom was choppy half an hour ago. You run a speed test from their laptop now and it reports high throughput and low latency. Does this show the network was not the cause? ○ Yes ● No

Why or why not?

(Answer inside the box)

the speed test is active and it is unclear what exactly is measured (eg. access link / wi-fi / device). It is also different from real user experience. So network could still have been the issue, eg. maybe the student was experiencing latency due to heavy loads.

7. [3 points]: (a) Name the five header fields that define a flow. In Hands-On 5 it was safe to group packets by only four of them; which one could be left out, and why? (b) Two packets match on all five fields but arrive 45 seconds apart with no packets in between. Would a router using conventional settings put them in the same flow record? Why or why not?

(Answer inside the box)

(a) Source IP, dest IP, source port, dest port, protocol. Ignored protocol because it was uniform.

(b) They would be separate since a default 30 s inactivity timeout would end the flow

Features and Representation

8. [4 points]: In Hands-On 6 you used `netml` to turn a capture into per-flow STATS features. nPrint instead turns each packet into a fixed-width bitmap of its header bits and lets a model learn which bits matter. Give **one advantage** and **one disadvantage** of *each* approach.

(Answer inside the box)

netml: advantage is it is interpretable and uses domain knowledge, disadvantage is it is not reproducible and doesn't transfer between tasks

nPrint: advantage is it is reusable across tasks and reproducible, disadvantage is it needs a lot of data and does not use timing or rates

9. [2 points]: A flow's protocol is TCP, UDP, or ICMP, and you encode it as 0, 1, 2. Which statements are true? Select all that apply.

- ● This is usually fine for a random forest
- ● A linear model treats ICMP as "twice" UDP
- ● One-hot encoding gives three binary columns
- ○ Min-max scaling fixes it for a linear model
- ○ Tree models require one-hot inputs
- ○ The order 0, 1, 2 carries real meaning

10. [3 points]: Some rows in your flow dataset have missing values. Give **one** example from network data where filling in the column average is reasonable, and **one** where it is a bad idea. Explain why for the second.

(Answer inside the box)

It is reasonable to fill in an ACK timestamp because it is bounded by its neighbors

It is bad to fill in a missing value which may be the signal itself of an attack, eg. a malformed header, since that would hide the signal

Initials: _____

Training and Evaluation

11. [3 points]: You have six months of hourly traffic volumes and want to forecast next month's load to decide whether to add a server. You shuffle all rows randomly, split 80/20, and get low test error. Is that a trustworthy estimate of how well the model will forecast? ○ Yes ● No

Why or why not?

(Answer inside the box)

Temporal data, so shuffling before split means we are training on future data - should just split, eg. train on earlier data and test on later data.

12. [5 points]: A detector labels 1,000 flows; 20 of them are attacks (attack is the positive class). Its confusion matrix:

	Predicted attack	Predicted benign
True attack	15	5
True benign	15	965

(a) Compute accuracy, precision, and recall. (b) What is the accuracy of a classifier that always says "benign"? What does the comparison tell you, and which curve would you use to compare detectors on this data?

(Answer inside the box)

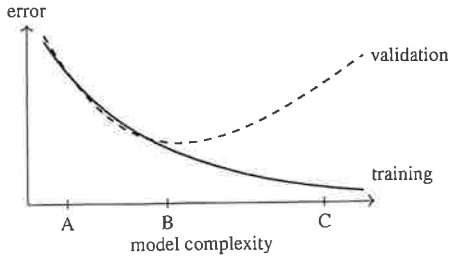
(a) Precision $= \frac{TP}{TP+FP} = \frac{15}{15+15} = \frac{15}{30} = \frac{1}{2}$ Recall $= \frac{TP}{TP+FN} = \frac{15}{15+5} = \frac{15}{20} = \frac{3}{4}$

Accuracy $= \frac{TP+TN}{TP+TN+FP+FN} = \frac{15+965}{1000} = \frac{980}{1000} = \frac{98}{100}$

(b) A classifier that always says benign predicts the 980 true benign correctly and the 20 true attack incorrectly, so it would have 98% accuracy. Since the classes are imbalanced, should use PR-curve.

↳ same as part (a)

13. [3 points]: The figure shows training and validation error as model complexity increases.



(a) Which point (A, B, C) has high bias? (b) Which is overfitting? (c) Name one way to reduce overfitting other than choosing a simpler model.

(Answer inside the box)

(a) A is high bias (b) C is overfitting

(c) More data, or early stopping (eg. limiting number of training epochs)

Initials: ____

Assignment 1: Video Quality Inference

14. [3 points]: In Hands-On 3 and Assignment 1 you computed a segment-download-rate feature from encrypted Netflix traffic. (a) How can you find segment boundaries without decrypting anything? (b) Why do segments carry information about resolution?

(Answer inside the box)

(a) Segments are bursts of packets, so they are separated by gaps (we used ≈1s gaps)
(b) Segments provide information on resolution based on the amount of data they carry, eg. higher bytes per segment may indicate greater resolution.

15. [4 points]: A student's Assignment 1 classifier scores 75% accuracy with a random train/test split, but about 30% when every session from one household is held out for testing. (a) What does this suggest the model learned? (b) An ISP wants to deploy the model in new homes. Which accuracy is the better estimate, and name one change that would help.

(Answer inside the box)

(a) This suggests the model learned the resolutions of specific homes based on signatures like RTT, device, and TCP behavior rather than inferring resolution through segment sizes / info, so it may not generalize well.
(b) 30% is more accurate because it reflects how the model will perform on new homes that aren't in the training set. A change that could help is dropping home_id and other columns that may differentiate households so that the model only uses good features.

Feedback

16. [3 points]: Interest (1=Boring!; 10=Amazing!): [8] Difficulty (1=Too easy; 10=Too hard): [5]

One thing you like. One suggestion for improvement:

(Answer inside the box)

I like the hands-ons. I don't have previous networking experience, so it has been a little difficult to keep up at times - more help on networking fundamentals could be useful.

Initials: BJ